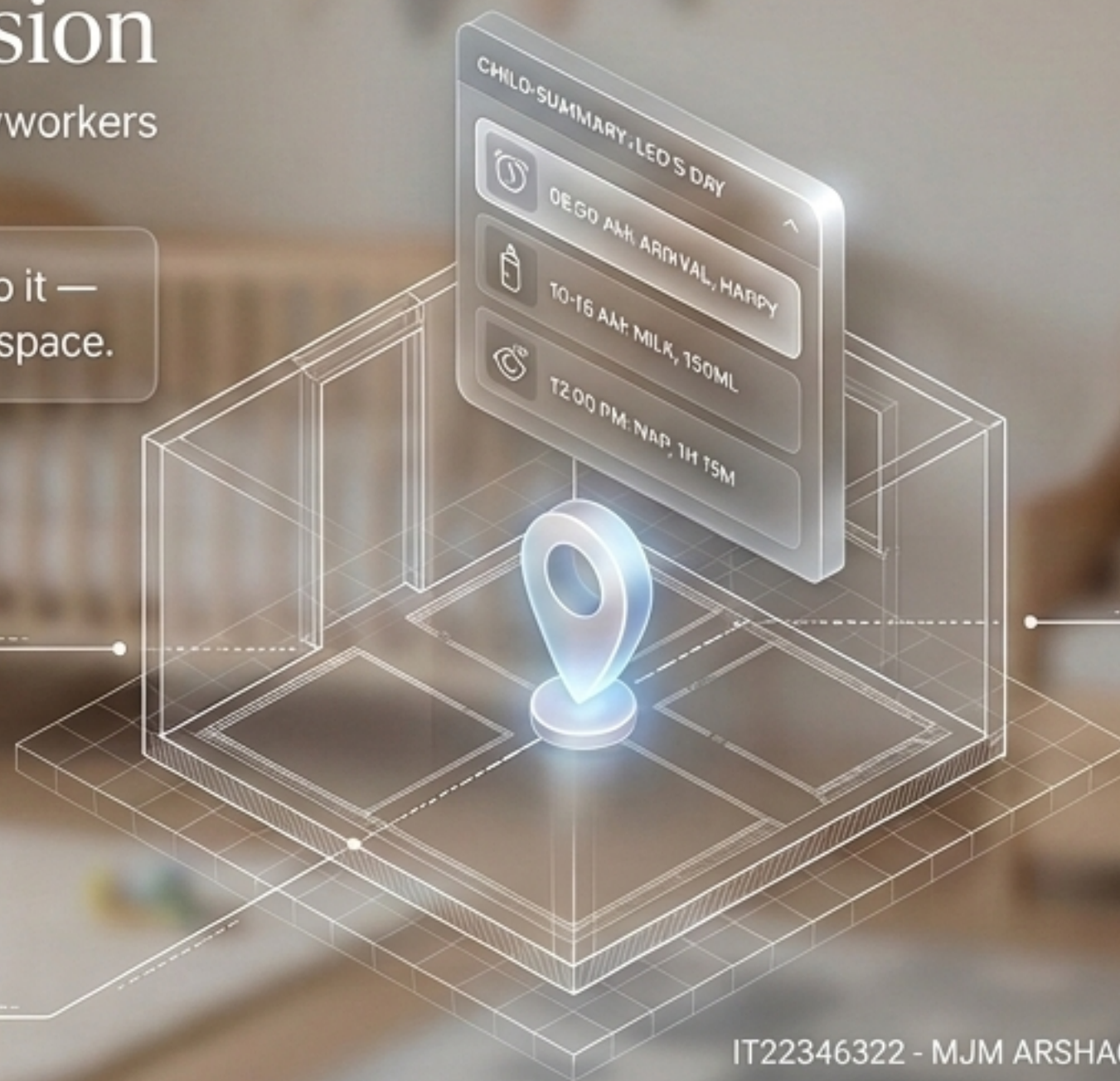


NurseryConnectVision

A Spatial Day Dashboard for Nursery Keyworkers

Glance at your whole room, then step into it —
tap a child to read their day, anchored in space.



CHILD SUMMARY: LEO'S DAY	
08:00 AM: ARRIVAL, HAPPY	
10:15 AM: MILK, 150ML	
12:00 PM: NAP, 1H 15M	

IT22346322 - MJM ARSHAQ

Case Study Module — Submission

The User



The Nursery Keyworker, responsible for managing a dynamic group of children.

The Problem

During busy handover times, workers need both an at-a-glance 'how is everyone' overview and fast, per-child detail.



⚠ The Physical Constraint

They are often moving around the room with their hands full. A traditional flat tablet buries crucial details behind taps and pulls the worker's gaze down to a screen, breaking visual contact with the children.



Flat iPad App



visionOS Dashboard

Posture

Flat List requires head-down scrolling.

Spatial Volume enables head-up, eyes-on-the-room monitoring.

Hands

Flat List requires physically holding a device.

Spatial Volume allows hands-free volume interaction via eye-tracking and micro-gestures.

Data Anchoring

Flat List buries data in nested menus.

Spatial Volume anchors detailed information directly next to the physical child/peg it describes in the 3D room.

Core Principle Insight: Only the part that earns depth becomes 3D.

Surface Plan

Window
(WindowGroup)

Volume
(.windowStyle(.volumetric))

Immersive Space



The roster list

A flat panel is the right tool for a scannable, text-dense list. No depth needed.

The programmatic nursery room

Depth adds value here through walk-around visibility and data anchored next to physical representation.

Out of scope

A bounded volume delivers the spatial payoff; full immersion adds user isolation and complexity with no extra benefit for the brief.

Step 1: Enter & Assess.

The user enters the app and views the text-dense flat panel for glanceable tracking.

UI Architecture

Features the roster list, the child-count toolbar, and the floating ornament.

The Bridge

The ornament serves as the spatial toggle and displays the current selection, acting as the gateway to the 3D room.

Children: 12/12 Staff: 2		+
	Anna B.	2y 3m
	Ben C.	1y 8m
	Chloe D.	3y 1m
	David E.	2y 6m
	Eva F.	1y 5m
	Frank G.	2y 9m
	Grace H. Henry I.	3y 0m
	Ivy J.	1y 7m
	Jack K.	2y 5m
	Leo L.	3y 2m
	Mia M.	1y 9m

 ☒ Room: Nursery A (3D Toggle)

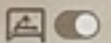
Elegant: spatial
"ornament"
horizontal toolbar.

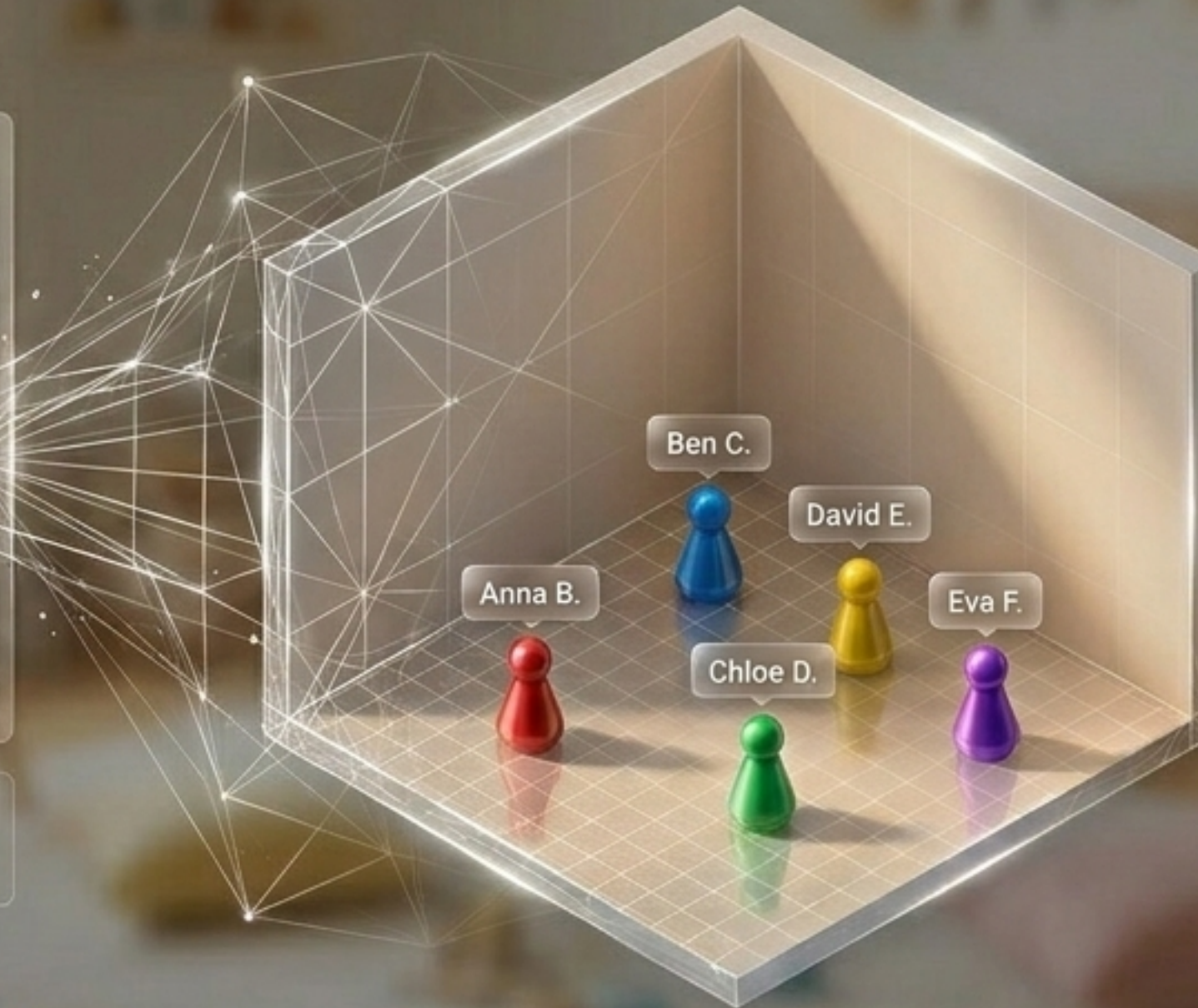
Step 2: Open Volume.

Toggling the view unfolds a programmatic nursery room consisting of a floor, two walls, and directional lighting.

Children: 12/12 | Staff: 2

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	Eva F.	1y 8m
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 Room: Nursery A (3D Toggle)



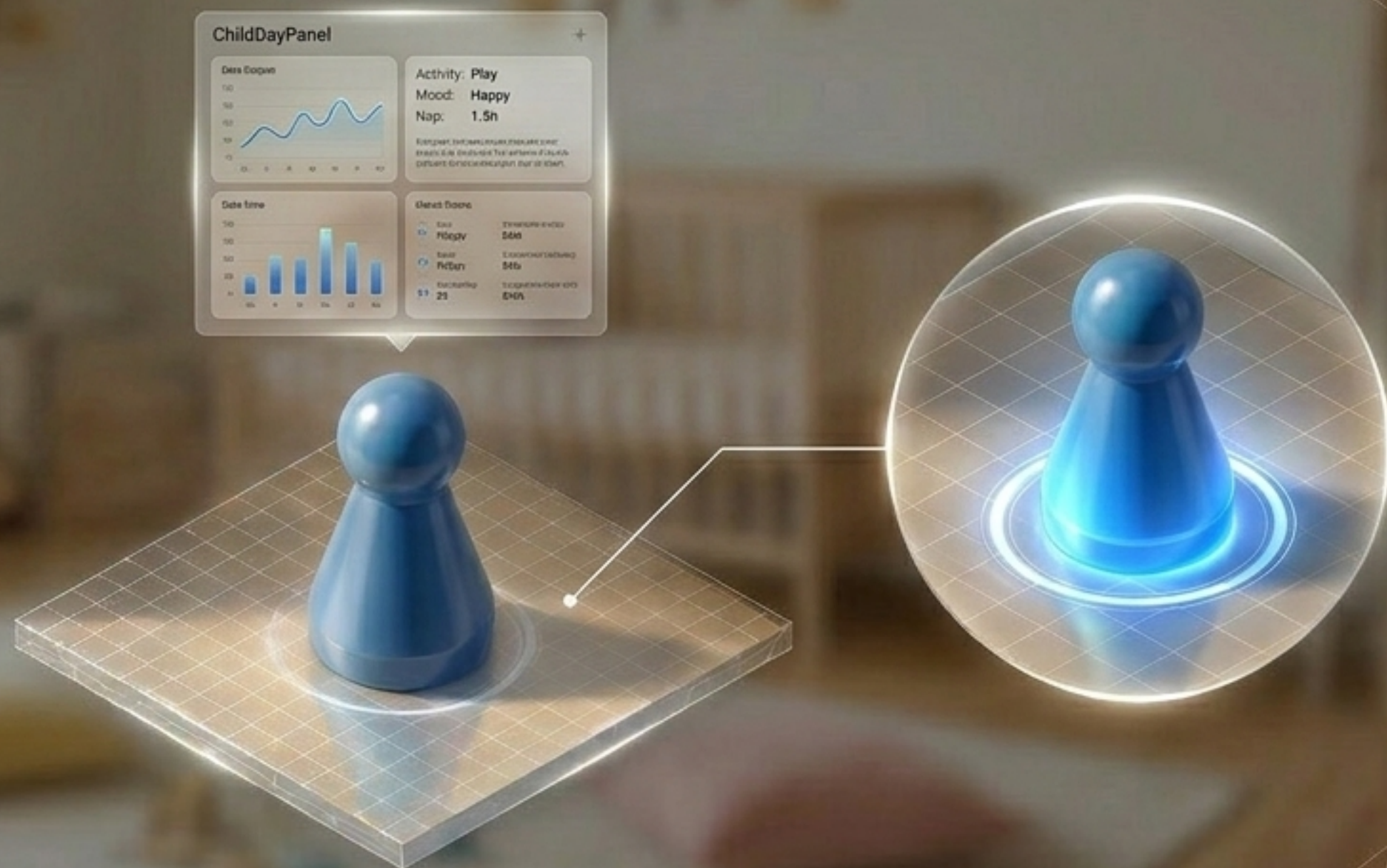
Data Representation

5 named pegs appear in the bounded 3D space, each representing a specific child in the physical room.

Hover: Looking at the peg triggers a highlight, providing immediate system feedback via `HoverEffectComponent`.

Tap: Pinching selects the peg, causing the `ChildDayPanel` to float out and anchor directly above it.

Cognitive Benefit: Information is spatially associated with the physical location of the object, eliminating the need to cross-reference a separate list.





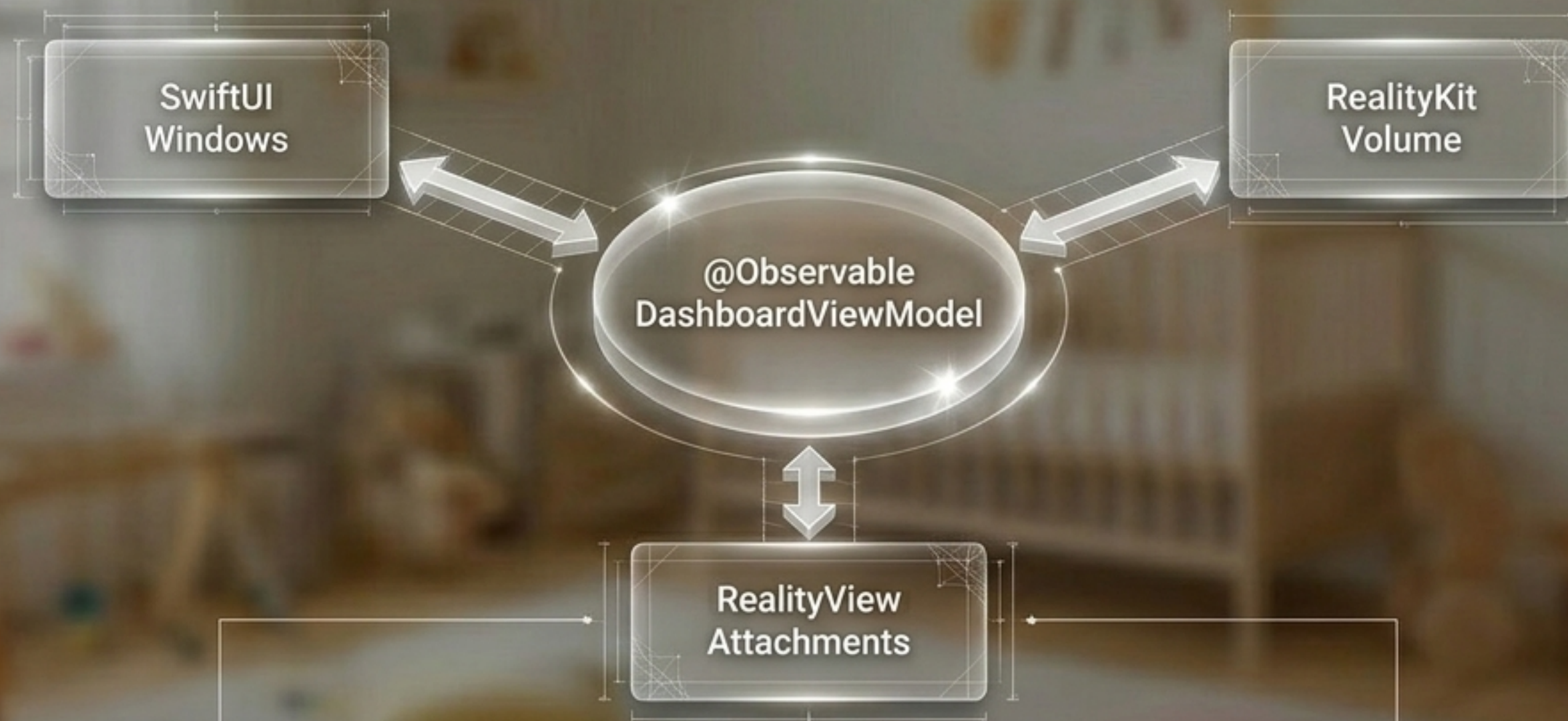
Drag to Rotate

Users can grab and rotate the entire room volume to view occluded pegs, enabling walk-around capability without moving their physical feet.



Spatial Confirmation Chime

A ~0.3s *chime.wav* plays upon tapping. Critically, it emanates spatially from the tapped peg, giving the digital sound a physical location to confirm the action.



Single Source of Truth

The DashboardViewModel ensures that the roster row highlight, the ornament label, and the 3D panel all instantly sync across dimensions.

Programmatic 3D Execution

The environment is built entirely in code via NurseryRoomBuilder (generating meshes, materials, and collisions)—bypassing Reality Composer Pro for complete programmatic control.